

Question 3

Find the point on the circle

$$x^2 + y^2 = 4$$

that is closest to the point $(3, 4)$.

Solution using Lagrange Multipliers

Step 1: Write the function to minimize

The distance between (x, y) and $(3, 4)$ is

$$D = \sqrt{(x - 3)^2 + (y - 4)^2}$$

Instead of minimizing D , we minimize D^2 :

$$f(x, y) = (x - 3)^2 + (y - 4)^2$$

Our constraint is

$$g(x, y) = x^2 + y^2 = 4$$

Step 2: Find the partial derivatives

For f :

$$f_x = 2(x - 3)$$

$$f_y = 2(y - 4)$$

For g :

$$g_x = 2x$$

$$g_y = 2y$$

Using Lagrange multipliers:

$$\nabla f = \lambda \nabla g$$

Therefore,

$$2(x - 3) = \lambda(2x)$$

$$2(y - 4) = \lambda(2y)$$

Simplifying:

$$x - 3 = \lambda x \quad (1)$$

$$y - 4 = \lambda y \quad (2)$$

Step 3: Eliminate λ

From (1):

$$x - 3 = \lambda x$$

$$\lambda = \frac{x - 3}{x}$$

From (2):

$$\lambda = \frac{y - 4}{y}$$

Set them equal:

$$\frac{x - 3}{x} = \frac{y - 4}{y}$$

Cross multiply:

$$y(x - 3) = x(y - 4)$$

$$xy - 3y = xy - 4x$$

$$4x = 3y$$

Thus,

$$y = \frac{4}{3}x$$

Step 4: Use the constraint

We know

$$x^2 + y^2 = 4$$

Substitute $y = \frac{4}{3}x$:

$$x^2 + \left(\frac{4}{3}x\right)^2 = 4$$

$$x^2 + \frac{16}{9}x^2 = 4$$

$$\frac{25}{9}x^2 = 4$$

$$x^2 = \frac{36}{25}$$

So,

$$x = \pm \frac{6}{5}$$

Since $y = \frac{4}{3}x$,

$$y = \pm \frac{8}{5}$$

The two points on the circle are

$$\left(\frac{6}{5}, \frac{8}{5}\right)$$

and

$$\left(-\frac{6}{5}, -\frac{8}{5}\right)$$

The point $(3, 4)$ lies in the **first quadrant**, so the closest point is the one in the same direction:

$$\left(\frac{6}{5}, \frac{8}{5}\right)$$

 **Final Answer**

$$\left(\frac{6}{5}, \frac{8}{5}\right)$$

is the point on the circle $x^2 + y^2 = 4$ closest to $(3, 4)$.